Yufeng Gu

4844 Bob and Betty Beyster Building, 2260 Hayward
Ann Arbor, MI, USA, 48109-2121
yufenggu@umich.edu +1-734-272-5271

EDUCATION

University of Michigan — **Ann Arbor, MI, USA**
Ph.D. Student, Computer Science and Engineering Department — *Sep. 2020 - Dec. 2025*
Advisor: Prof. Reetuparna Das, GPA: 3.88/4.00

Zhejiang University (ZJU) — **Hangzhou, China**
B.Eng., College of Information Science and Electronic Engineering — *Sep. 2016 - June 2020*
GPA: 3.96/4.00

PROFESIONAL EXPERIENCE

University of Michigan — **Ann Arbor, MI, USA**
Graduate Research Assistant, Advisor: Reetuparna Das — *Sep. 2020 – Dec. 2025*
Optimized systems and architectures for emerging applications, including Generative AI and genome sequencing analysis. Proposed solutions across domain specific accelerator, near-memory processing and QoS-aware hardware resource scheduler.

Tenstorrent Inc. — **Santa Clara, CA, USA**
Performance Architect Intern, Manager: Wei-han Lien — *May 2023 – Aug. 2023*
Built performance model for RISC-V CPU and AI accelerator.

Intel Labs — **Hillsboro, OR, USA**
Graduate Research Intern, Manager: Nilesh Jain — *June 2022 – Aug. 2022*
Optimized AI QoS on next generation heterogenous datacenter platform.

Yale University — **New Heaven, CT, USA**
Undergraduate Research Intern, Advisor: James Duncan, Xiaoxiao Li — *Nov. 2019 – Mar. 2020*
Interpretation on ASD with fMRI and Deep Learning Models.

École Polytechnique Fédérale De Lausanne (EPFL) — **Lausanne, Switzerland**
Undergraduate Research Intern, Advisor: Babak Falsafi — *July 2019 – Sep. 2019*
Patched QFlex Simulator and accelerated activation functions for DNN.

PEER-REVIEWED CONFERENCE/JOURNAL PUBLICATIONS

* equal contribution

Yufeng Gu, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." (Invited Paper) Communications of the ACM, 2025.

Alireza Khadem*, Kamalavasan Kamalakkannan*, Zhenyan Zhu, Akash Poptani, **Yufeng Gu**, Jered Benjamin Dominguez-Trujillo, Nishil Talati, Daichi Fujiki, Scott Mahlke, Galen Shipman, Reetuparna Das. "DX100: Programmable Data Access Accelerator for Indirection." In Proceedings of the 52th Annual International Symposium on Computer Architecture (ISCA'25).

Yufeng Gu*, Alireza Khadem*, Sumanth Umesh, Ning Liang, Xavier Servot, Onur Mutlu, Ravishankar Iyer, Reetuparna Das. "DX100: Programmable Data Access Accelerator for Indirection." In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25).

Alireza Khadem, Daichi Fujiki, Hilbert Chen, **Yufeng Gu**, Nishil Talati, Scott Mahlke, Reetuparna Das. "Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing." In 2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA'25). **Distinguished Artifact Honorable Mention (3/29)**

Yufeng Gu, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." In Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA'23). **Communication of ACM Research Highlights (24/10,000+)**

Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. "GenomicsBench: A Benchmark Suite for Genomics." In IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS'21).

Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. "Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results." Medical Image Analysis 65: 101765. *IF = 11.1*

Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. "Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty." In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI'20).

UNDER REVIEW CONFERENCE SUBMISSIONS

Data Reuse-Aware PIM Architecture for Efficient LLM Inference. *Under submission (First Author)*

Fast Resource Management for QoS-Aware Co-Location of ML Inference. *Under submission (First Author)*

WORKSHOP PUBLICATIONS

Noah Kaplan, **Yufeng Gu**, Reetuparna Das. "Understanding and Characterization of Pangenomics." in International Workshop on Domain Specific System Architecture (DOSSA) co-located with ISCA 2024.

AWARDS & HONORS

Distinguished Artifact Honorable Mention (HPCA 2025) *March 2025*
MultiDimensional Vector ISA Extension for Mobile In-Cache Computing. (3/29 Artifacts)

HPCA Travel Grant *March 2025*

Communucation of ACM (CACM) Research Highlights *July 2024*
GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis. *CACM Research Highlights section selects 24/10,000+ papers from ACM conferences per year, reprinting the most significant and influential results across Computer Science.*

University of Michigan Rackham Graduate Student Research Grant ($3000) *Apr. 2024*

ISCA Travel Grant *July 2023*

University of Michigan Rackham Travel Grant *2023, 2025*

Fellowship of Summer@EPFL (2% applicants awarded) *July 2019*

Tang Lixin Fellowship (2‰ students awarded) *Nov. 2017, 2018, 2019*

Outstanding Student Leaders in Zhejiang University (3% students awarded) *Oct. 2017, 2019*

First-Class Scholarship for Outstanding Students (2% students awarded) *Oct. 2017*

TECHNICAL SKILLS

Programming: C/C++, Python, CUDA, Verilog, Bash

Tools: Linux, Docker, Git, Slurm, Synopsis Design Compiler

Framework: PyTorch, vLLM

SERVICES

Artifact evaluation reviewer for IISWC 2021, ISCA 2023, 2024, 2025.

Reviewer for University of Michigan graduate student admission *Jan. 2021, 2022*

Voluntary teacher in Tuanlin Primary School, Guizhou, China *Aug. 2017*

MENTORSHIP

Dev Singhania M.S. UM	*Jan. 2025 - Apr. 2025*
Zesen Zhao M.S. UM	*Jan. 2025 - Aug. 2025*
Wenjie Geng M.S. UM	*Sep. 2024 - Aug. 2025*
Mayne Mei B.Eng. UM	*Jan. 2024 - Apr. 2024*
Ao Luo M.S. UM	*Jan. 2024 - June 2024*
Yuzhe Ruan B.S. UM Yale	*Sep. 2023 - Apr. 2024*
Donglin Yu B.Eng. UM UIUC	*Jan. 2023 - Apr. 2024*
Dawit Melka B.Eng. Addis Ababa University	*May 2022 - Aug. 2022*
James Gu B.Eng. UM	*May 2022 - Aug. 2022*